Lecture 5 $\boxed{\text{Continue...}}$

Inverse Trigonometric Function

Formulas:

1. $\frac{d}{dx}[\sin^{-1} u] = \frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$, $\frac{d}{dx}[\cos u] = -\frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$

2. $\frac{d}{dx}[\tan^{-1} u] = \frac{1}{1+u^2} \frac{du}{dx}$, $\frac{d}{dx}[\cot^{-1} u] = -\frac{1}{1+u^2} \frac{du}{dx}$

3. $\frac{d}{dx}[\sec^{-1} u] = \frac{1}{|u|\sqrt{u^2-1}} \frac{du}{dx}$, $\frac{d}{dx}[\csc^{-1} u] = -\frac{1}{|u|\sqrt{u^2-1}} \frac{du}{dx}$

e.g. $y = \sin^{-1}(x^3)$

$$\frac{dy}{dx} = \frac{d}{dx} \sin^{-1}(x^3)$$

$$= \frac{1}{\sqrt{1-(x^3)^2}} \cdot \frac{d}{dx}(x^3)$$

$$= \frac{3x^2}{\sqrt{1-x^6}} \quad \ast$$

Extra Problem: Find $\frac{dy}{dx}$.

1. $y = \ln(1 - xe^{-x})$
2. $y = \pi^{\sin x}$
3. $y = x^{\sin x}$
4. $y = 4^{3\sin x - e^x}$
5. $y = 2^{\cos^{-1} x^2 + \ln x}$
6. $y = \tan^{-1}(x^3)$
7. $y = e^x \sec^{-1} x$
8. $y = x^2 (\sin^{-1} x)^3$

page: 202

Problems: 39-42, 44-51

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Le - masculine

La - feminine

L'Hôpital's Rule

Earlier we ~~were~~ discussed only able to conjecture using numerical or graphical evidence for establishing the limits.

Now we use one theorem which is an extremely powerful tool that is used internally by many computer programs to calculate limits of various types.

Theorem (L'Hôpital's Rule for $\frac{0}{0}$ form)

Suppose that f and g are differentiable functions on an open interval containing $x=a$, except possibly at $x=a$, and that

$$\lim_{x\to a} f(x)=0 \text{ and } \lim_{x\to a} g(x)=0.$$

If $\lim_{x\to a}[f'(x)/g'(x)]$ exists, or if this limit is $+\infty$ or $-\infty$, then

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}$$

$\boxed{\text{Same for } \frac{\infty}{\infty} \text{ form}}$ $\lim_{x\to a} f(x)=\infty$ and $\lim_{x\to a} g(x)=\infty$.

If $\lim_{x\to a}[f'(x)/g'(x)]$ exists, then

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}.$$

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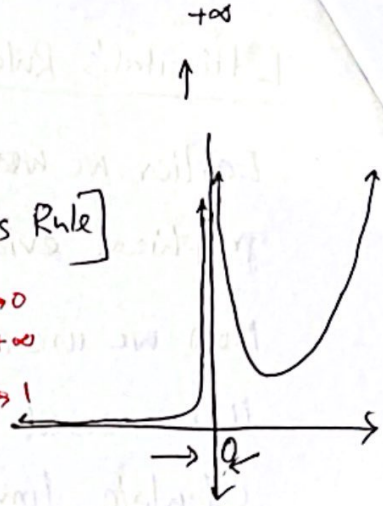
Example: $\lim_{x \to 0} \frac{e^x - 1}{x^3}$ $\quad \frac{0}{0}$

$= \lim_{x \to 0} \frac{e^x}{3x^2}$ [Apply L'Hopital's Rule]

$\frac{1}{x^2}$ is even, so when $x \to 0$ $\frac{1}{x^2} \to +\infty$

$= +\infty$

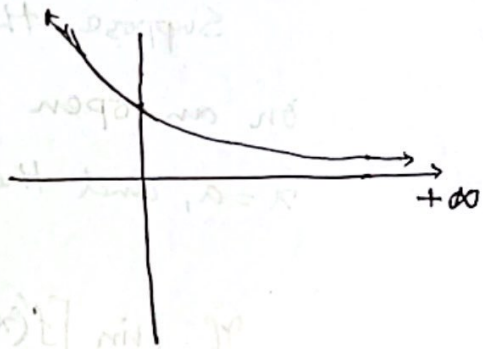
$e^x > 0$ when $x \to 0$ $e^x \to 1$



Example: $\lim_{x \to +\infty} \frac{x}{e^x}$ $\quad \left[\frac{\infty}{\infty}\right]$

$= \lim_{x \to +\infty} \frac{1}{e^x}$ [Applying L'Hopital's Rule]

$= 0$



* $0 \cdot \infty$ ~~also~~ in.

$0 \cdot \infty$ is also another indeterminate form. ~~But~~ ~~How~~ But it gives us sometime wrong result.

Since "zero times anything is zero". However, this is fallacious since $0 \cdot \infty$ is not a product of numbers, but rather a statement about limits.

So, when we get $0 \cdot \infty$, we will try to make it $\frac{\infty}{\infty}$ or $\frac{0}{0}$ form.

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Example: $\lim_{x \to \pi/4} (1-\tan x)\cdot \sec 2x$ $\quad 0\cdot\infty$ form

So we make it like $\frac{0}{0}$

$$\therefore \lim_{x\to \pi/4} \cancel{\tan}(1-\tan x)\sec 2x = \lim_{x\to\pi/4} \frac{1-\tan x}{\frac{1}{\sec 2x}}$$

$$= \lim_{x\to\pi/4} \frac{1-\tan x}{\cos 2x} \qquad \frac{0}{0}$$

$$= \lim_{x\to\pi/4} \frac{-\sec^2 x}{-2\sin 2x}$$

$$= \frac{-2}{-2} = 1$$

Example: $\lim_{x\to 0}\left(\frac{1}{e^{1/2x}}\right)^x$; let $f(x) = \frac{1}{e^{1/2x}}$ $\quad g(x) = x$

$\lim_{x\to 0} f(x) = 0$ $\quad \lim_{x\to 0} g(x) = 0$

So it has 0^0 form.

Now $y = \left(\frac{1}{e^{1/2x}}\right)^x$

$$\ln y = x\ln\left(\frac{1}{e^{1/2x}}\right) = \cancel{x\ln} \; x\left[\ln 1 - \ln e^{1/2x}\right]$$

$$= x\left[0 - \frac{1}{2x}\ln e\right] = x\left(-\frac{1}{2x}\cdot 1\right)$$

$$= -\frac{1}{2}$$

$$\therefore \lim_{x\to 0} \ln y = \lim_{x\to 0}\left(-\frac{1}{2}\right)$$

$$= -\frac{1}{2}$$

$$y = e^{-1/2}$$

Problem: $\lim_{x\to\infty}\left(\frac{2}{x}\right)^{e^{-x}}$

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Problem: Evaluate $\lim_{x\to 0^+}\left(\frac{1}{x}-\frac{1}{\sin x}\right)$ $[\infty-\infty]$ form

[So we want to make it $\frac{0}{0}$ or $\frac{\infty}{\infty}$]

Now, $\lim_{x\to 0^+}\left(\frac{\sin x - x}{x\sin x}\right)$ $\left[\frac{0}{0}\text{ form}\right]$

$$= \lim_{x\to 0^+}\frac{\cos x - 1}{\sin x + x\cos x} \quad \left[\frac{0}{0}\right]$$

$$= \lim_{x\to 0^+}\frac{\sin x}{\cos x + \cos x - x\sin x}$$

$$= \lim_{x\to 0^+}\frac{\sin x}{2\cos x - x\sin x}$$

$$\boxed{= \frac{0}{2}} = 0$$

Problem: Prove that $\lim_{x\to 0^+}(1+x)^{1/x} = e$

Sol^n: Since this limit form 1^∞ form.

So, let $y = (1+x)^{1/x}$

$$\ln y = \ln(1+x)^{1/x} = \frac{1}{x}\ln(1+x)$$

$$\lim_{x\to 0}\ln y = \lim_{x\to 0}\frac{\ln(1+x)}{x} \quad \left[\frac{0}{0}\text{ form}\right]$$

$$= \lim_{x\to 0}\frac{1/(1+x)}{1}$$

$$= \lim_{x\to 0}\frac{1}{1+x} = 1$$

So, $\ln y = 1 \Rightarrow y = e^1 = e$ i.e. $\lim_{x\to 0}(1+x)^{1/x} = e$

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Some Problems:

Page : 227 Book : Reference book - H. Anton

~~Problems:~~

Exercise : 3.6

Problems: 7, 27, 29, 34, 36, 37,

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